Pandit Deendayal Upadhyaya Innovation & Incubation Centre
Guru Jambheshwar University of Science and Technology,
Hisar, Haryana 125001

Seed Funding

Project Details

1. **Title of Startup Idea/ Project:** Velo – Smart Campus Electric Mobility Platform

VELO

Go Smart. From the Start.

2. **Date of Submission:** 10-07-2026

3. **Team Leader:**

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5. **Faculty Mentor: Dr. Sunil Verma**

6. **Duration of Project: 6 Month**

7. **Estimate of Budget: 1.5 Lakh**

8. Project at a Glance:

8.1 Brief Description of the Idea: Velo is a smart campus mobility platform designed to provide students with affordable, convenient, and eco-friendly transportation through electric scooters and bicycles. Using a mobile application, students can rent vehicles on an hourly or daily basis for short-distance travel to destinations such as markets, railway stations, bus stands, hospitals, coaching centres, and nearby commercial areas.

8.2 Domain:

- Electric Mobility (EV)
- Bicycle Sharing
- Smart Transportation
- Mobility-as-a-Service (MaaS)
- Shared Economy
- Sustainable Mobility

8.3 Problem Identified: Students residing in hostels and day scholars frequently face challenges in accessing affordable and reliable transportation for short-distance travel. Existing options such as auto-rickshaws and cabs are often expensive, involve long waiting times, and are not always available during peak hours or emergencies. Additionally, many campuses lack dedicated mobility solutions, making commuting inconvenient. Students also hesitate to purchase personal bicycles or scooters due to cost, maintenance, parking, and limited usage.

8.4 Proposed Solution: Develop a campus-focused mobility platform that enables verified students to rent bicycles and electric scooters through a mobile application. The platform will offer hourly and daily rental plans, secure digital verification, GPS-enabled tracking (for electric scooters), digital payments, and a seamless booking experience.

8.5 Target Beneficiaries / Users:

- University and college students
- Hostel residents
- Day scholars
- Research scholars
- Faculty and staff (future phase)
- Educational institutions seeking sustainable campus mobility solutions

8.6 Key Innovation:

Velo is a campus-first shared mobility platform offering bicycles and electric scooters for students.

- The MVP will start with 2 bicycles and 2 electric scooters to test demand and collect user feedback.
- Students can rent vehicles on an hourly or daily basis.
- The platform will provide secure student verification and digital payments.
- Users can book vehicles through a simple technology-enabled booking system.
- Velo aims to provide affordable, convenient, and sustainable transportation for campus commuting.
- After validating the model, the platform can expand to universities, corporate campuses, tech parks, and gated communities across India.

8.7 Current Stage of Development:

- Idea conceptualization completed.
- Team formation completed.
- Initial market research initiated.
- Student demand survey launched.
- Business model and pricing validation in progress.
- MVP deVelopment planned after validation and funding.

8.8 Expected Impact:

- Reduce transportation costs and travel time for students.
- Improve last-mile connectivity within and around university campuses.
- Promote cycling and electric mobility as sustainable alternatives to fuel-based transportation.
- Reduce traffic congestion and carbon emissions in campus areas.
- Establish a scalable campus mobility solution that can be replicated across educational institutions throughout India.
- Generate entrepreneurial and employment opportunities while contributing to greener and smarter campuses.

9. Declaration:

- I hereby declare that the information provided by our team is true and authentic, and we shall abide by all rules and regulations of the program/institution.(Team Leader)
- I hereby certify that the proposed idea/project has been reviewed by me and I agree to guide and mentor the student team throughout the program.(Mentor)
- I hereby certify that the above-mentioned students are bona fide students of the department, and the department recommends their participation in the program.(Head/Chairperson)

Sneha 10/07/26 | 10/7/26 | 10/7/26

Team Leader (Sign.) | Mentor (Sign). | Head/Chairperson (Sign.)

Detailed Project Proposal

Executive Summary

Velo is a student-led smart mobility startup focused on providing affordable, convenient, and sustainable transportation within university campuses. The platform enables students to rent bicycles and electric scooters through a mobile application on an hourly or daily basis. The project aims to solve the challenge of limited and expensive last-mile transportation by offering a shared mobility solution tailored specifically for educational institutions.

The startup will begin with a pilot fleet of two bicycles and two electric scooters at Guru Jambheshwar University of Science & Technology (GJUST). Through this pilot, the team will validate demand, optimize operations, and refine the business model before expanding to other universities. By leveraging technology, digital verification, and smart booking, Velo seeks to make campus transportation more accessible while promoting sustainable mobility.

Introduction of the Startup

Students frequently travel between hostels, academic blocks, markets, railway stations, bus stands, hospitals, coaching centres, and other nearby destinations. However, existing transportation options such as auto-rickshaws and cabs are often expensive, inconvenient, or unavailable during peak hours. Many students also do not own personal vehicles due to the associated costs of purchasing, maintenance, and parking.

Velo aims to bridge this gap by introducing a shared mobility platform that provides bicycles and electric scooters on a rental basis. Through a dedicated mobile application, verified students can easily locate, book, unlock, and rent vehicles whenever required. The platform focuses on affordability, convenience, safety, and sustainability while creating a modern transportation ecosystem within educational campuses.

The long-term vision is to expand beyond universities and establish Velo as a leading shared mobility platform for campuses, corporate parks, residential communities, and smart cities across India.

Problem Statement

Transportation remains one of the most common challenges faced by university students, particularly hostel residents. Students frequently require short-distance transportation for essential activities, but the available options are often inefficient and expensive.

The key challenges include:

- High dependence on auto-rickshaws and cab services for short-distance travel.
- High transportation costs for students with limited budgets.
- Limited availability of transport during early mornings, late evenings, and peak hours.
- Lack of dedicated mobility services within and around university campuses.
- Inconvenience of owning and maintaining a personal bicycle or scooter.
- Increasing traffic congestion and carbon emissions due to excessive use of fuel-powered

vehicles.

- Limited adoption of sustainable transportation solutions within educational institutions.

These challenges negatively impact students' time, convenience, and overall campus experience.

Proposed Solution

Velo proposes a **technology-enabled shared mobility platform** that allows verified students to rent bicycles and electric scooters through a user-friendly mobile application. Users will be able to view vehicle availability, reserve a vehicle, make digital payments, and access the vehicle for their journey through a seamless booking process.

As part of the **Minimum Viable Product (MVP)**, the startup will deploy a fleet of **two bicycles and two electric scooters** to evaluate user demand, operational efficiency, pricing strategies, and customer experience within the university campus. This phased approach minimizes initial investment while generating valuable insights for future expansion.

Unlike conventional city-wide rental services, Velo is designed specifically for campus ecosystems, focusing on student affordability, convenience, and safety. The platform combines sustainable transportation with digital technology to create a scalable mobility solution that can be replicated across educational institutions, corporate campuses, gated communities, and other closed environments throughout India.

Objectives of the Project

Primary Objective: To establish a sustainable, affordable, and technology-driven electric scooter rental service exclusively for university students, providing convenient transportation across the campus while promoting eco-friendly mobility.

Specific Objectives:

1. Reduce travel time between hostels, academic blocks, libraries, cafeterias, sports complexes, and other campus facilities.
2. Provide an affordable transportation alternative for students.
3. Promote green mobility through electric vehicles.
4. Improve students' productivity by minimizing travel time.
5. Develop an app-based rental system with QR-code unlocking, GPS tracking, and digital payments.
6. Ensure safety through speed limits, GPS monitoring, and responsible usage policies.
7. Build a scalable startup model that can expand to other universities after successful implementation.

Vision and Mission

Vision:

To become the leading smart campus mobility solution by providing affordable, sustainable, and technology-enabled electric transportation for university students.

Mission:

- Provide safe, reliable, and convenient electric scooter rentals for students.
- Promote eco-friendly transportation and reduce carbon emissions.
- Deliver a seamless digital experience through a mobile application.

• Support the deVelopment of smart and sustainable university campuses.

Market Analysis & Opportunity

Market Need: Large university campuses require students to travel long distances every day between hostels, classrooms, libraries, laboratories, cafeterias, and sports facilities. Walking consumes time and energy, creating demand for a quick, affordable, and eco-friendly transport solution. Market Opportunity: Thousands of students commute daily within the campus. Even if only 15–20% of students use the service regularly, it represents a strong and sustainable business opportunity. Growing awareness of electric vehicles and smart-campus initiatives further supports this startup.

SWOT Analysis:

Strengths: Eco-friendly, affordable, convenient, app-based service, low operating cost.
Weaknesses: Initial investment, battery replacement, maintenance.
Opportunities: Expansion to other universities, subscription plans, partnerships, advertising.
Threats: Scooter damage, theft, competition from personal vehicles, seasonal demand.

Target Customers

Primary Target Customers: University students residing in hostels or commuting within the campus.

Student Beneficiaries:

- Faster travel across campus
- Reduced physical fatigue
- Better punctuality for lectures and activities
- Affordable transportation
- Improved campus experience Environmental Benefits:
- Reduced carbon emissions
- Lower noise pollution
- Cleaner and greener campus environment

Value Proposition: "Our electric scooter rental service offers students a fast, affordable, safe, and eco-friendly way to travel across the university campus through a simple mobile application."

Business Model

Our startup provides electric scooters on rent exclusively for university campuses. Students can book scooters through a mobile app or WhatsApp and pay on an hourly, daily, or monthly basis.

Revenue Sources

- Hourly rentals
- Daily rentals
- Monthly subscription plans
- Security deposits
- Late return charges
- University and hostel partnerships (future)

Business Strategy

- Launch with 2 electric scooters and 2 bicycle at GJUST.
- Validate demand through a pilot program.
- Expand the fleet based on customer feedback and usage.
- Scale the model to universities across India.

Service Description

Service Name: Velo is a smart electric scooter rental platform designed specifically for university students.

Key Features

- Easy booking through a mobile app or WhatsApp
- QR code/OTP-based scooter access
- GPS tracking
- College ID verification
- Digital payments
- Fixed pickup and return locations
- Helmet provided with every scooter
- Customer support

Target Users

- Hostel students
- Day scholars

Technology & Innovation Used

The platform uses modern technology to provide a safe and convenient mobility experience.

Technologies Used

- Mobile application for bookings
- GPS tracking for real-time vehicle monitoring
- QR code/OTP-based scooter unlocking
- UPI-based digital payment integration
- Cloud database for user and ride management
- Analytics dashboard for fleet monitoring

Future Innovations

- Smart IoT locks
- AI-based demand prediction
- Automated fleet management
- Predictive maintenance system

Competitive Analysis

Feature	Our Startup	Yulu	Bounce	Veo
Primary Focus	University Campuses	City Mobility	City Mobility	University Campuses (USA)
Target Users	Students	General Public	General Public	Students
Campus-Specific Service	Yes	No	No	Yes
India Presence	Yes	Yes	Yes	No
Hourly Rental	Yes	Yes	Yes	Yes
Monthly Plans	Yes	Yes	Limited	Yes
University Partnerships	Yes	No	No	Yes

Marketing & Growth Strategy

What the MVP Actually Is

The MVP for this venture is not a scooter fleet — it is a small, cheap experiment designed to answer three questions before real money is committed: will students actually pay for a ride that is faster than walking and cheaper than an auto; can a 2-4 vehicle fleet be kept charged, clean, and available without full-time paid staff; and does demand cluster predictably enough (by time, by route) that a larger fleet could run profitably.

Concretely, the MVP is 2-4 vehicles, one fixed pickup/return stand close to where students already are, and a booking process that runs on WhatsApp or a Google Form rather than a custom app. A full mobile application, multiple campus stands, and a large fleet are Phase 2-4 outcomes, not Day-1 requirements — this mirrors how Yulu, Bounce, and Vogo all began with a few hundred vehicles in a single Bengaluru neighbourhood before expanding city-wide.

The survey gives a direct justification for testing rather than skipping to scale: 80.2% of 116 respondents said they have faced difficulty finding transport, and 69.8% said they would definitely (41.4%) or probably (28.4%) use a campus e-scooter rental if it existed. That is a strong enough signal to justify a pilot — but it is stated interest, not committed rupees, which is exactly what an MVP is built to test cheaply.

Evidence Base: What the Survey Actually Shows

Every marketing decision below is tied to a specific finding from the 116-response campus survey rather than an assumed number.

Survey Finding	Marketing / Strategy Implication
80.2% have faced difficulty finding transport; top complaints were long waiting time (67.2%) and auto being expensive (42.2%).	Core value proposition leads with "no waiting, lower cost than auto" — not novelty or sustainability messaging.
2nd-year (41.7%) and 3rd-year (40%) students make up 81.7% of respondents; 1st, 4th and 5th year are a small minority.	Primary target segment is 2nd/3rd-year students, not freshers.

43.5% live in the Girls Hostel, 36.5% in the Boys Hostel, 13.9% are day scholars.	Hostel residents are ~80% of demand. Stand #1 sits between the hostel cluster and the market/academic block — matching the direct respondent suggestion: "the scooter stand is close to the girls' hostel and the departments."
Current travel mode outside campus: Walk 40.9%, Auto 32.2%, Friend's Bike 11.3%, Own Vehicle 10.4%.	Walkers are the largest convertible segment; auto users are the most price-sensitive and already primed to pay for transport.
39.7% go outside campus daily and 31% go 2-3 times/week — about 71% are frequent travellers.	Large recurring-demand base supports a membership layer once reliability is proven (see 15.1).
Top destinations: Market (79.3%), Within University (69%), Bus Stand (58.6%), Restaurant/Cafe (44.8%).	Route priority: Hostel ↔ Market ↔ Bus Stand; hospital and coaching-centre routes (under 10% each) are Phase 3+ additions.
56.9% of respondents chose the ₹60-80/hour bracket, but open-text comments repeatedly said "Rent should be 30-50, as auto takes 20" and warned that high hourly pricing pushes students back to autos/Rapido.	Treat ₹60-80/hour as a ceiling, not a target: pilot a lower per-trip/per-minute price (15.2) to avoid the exact objection respondents raised.
Peak demand windows: 4-6 PM (64.9%), 12-2 PM (44.1%), 6-8 PM (38.7%).	Charging/rebalancing should happen outside these windows so availability is highest when demand peaks.
Recurring open-text requests: helmets provided, stands near hostel, per-minute rather than per-hour billing, ID-card-gated access.	Treated as MVP requirements, not later add-ons.

Marketing Objective

- Acquire 100-150 verified student users within the first 6-8 weeks of pilot launch.
- Reach 40%+ weekly repeat usage among registered users by the end of Phase 1.
- Validate a workable price inside the ₹20-50/trip range indicated by open-text feedback, rather than defaulting to the ₹60-80/hour bracket the same respondents pushed back on.
- Do this on a marketing spend of roughly ₹8,000-10,000 (16.2), by relying on campus-network channels instead of paid advertising.

Target Customer Segmentation

Segment	Approx. Share	Core Need	Primary Channel
Primary: 2nd/3rd-yr hostel residents	~65-70%	Fast, cheap, safe hostel ↔ market/academic-block trips, several times a week	Hostel notice boards, warden announcements, hostel WhatsApp groups
Secondary: Day scholars & PG residents	~14-20%	Reliable connectivity to bus stand/railway station for commute	Department noticeboards, bus-stand stand visibility
Tertiary: 4th/5th-yr & occasional users	~15%	Occasional trips to hospital, coaching centres, off-campus errands	Referral program, exam/placement-season posts

Value Proposition

Built directly from the gap respondents described between three existing options — walking (free but slow), auto (fast but expensive and unreliable at odd hours), and waiting (the single most-cited complaint at 67.2%): faster than walking and cheaper than an auto for trips under 3-4 km; no waiting, since vehicles are staged at a fixed stand rather than requiring a call or app-hailing wait; student-only, ID-gated access, directly answering the survey's outsider-use concern; and helmets provided as standard, addressing both the most frequent open-text request and the safety gap.

Customer Acquisition — First 100 Users

1. Direct re-engagement of the 115-116 survey respondents — a warm list, not cold outreach.
2. Founder-led sign-ups during hostel mess hours and department break times in the first two weeks.
3. A time-boxed "first 5 riders ride free" incentive for the opening week — the same tactic Uber used to onboard riders to Yulu.
4. Two to three volunteer campus coordinators (one per hostel block), paid in ride credits, not cash.

Offline Marketing

- Posters at hostel notice boards, mess entrances, and department corridors.
- Physical sign-up booth at the stand for the first two weeks.
- Demo/orientation slot during any university welcome event, subject to permission

Partnership Opportunities

Yulu's early growth leaned on a B2G2C approach — institutions provided space and internal distribution (for example, an office park giving it parking and emailing its own employees) rather than Yulu acquiring every user directly. The same logic applies here: university administration, hostel wardens, and the local market association are distribution partners, not just permission-givers.

- University administration: parking/stand space, permission to operate, inclusion in official student-welfare communication.
- Hostel wardens: announcement access and legitimacy with parents/students — important given the female-student safety comments in the survey.
- Local market association: a return/drop point at the market end of the route, reducing one-way vehicle imbalance.
- Phase 3-4: coaching centres and inter-departmental shuttle needs, as scalability options — not Phase 1 dependencies.

Retention Strategy

Since 67.2% of respondents identified long waiting time as the core problem, retention here is mostly an operations problem wearing a marketing hat: if the vehicle is reliably there, users come back.

- Reliability first: keep the Phase 1 fleet small enough (2-4 vehicles) to guarantee near-100% uptime.
- A simple punch-card reward (every 6th ride discounted) once repeat-usage data is stable.
- Post-ride WhatsApp feedback prompt to catch dissatisfaction before it becomes churn.
- A women-only vehicle slot/time-window, directly responding to safety-related survey comments.

Marketing-Relevant Risks

Risk	Likelihood	Impact	Mitigation
Hourly pricing (₹60-80) triggers the exact auto-comparison backlash seen in open-text comments.	High	High	Pilot at ₹20-50/trip; treat ₹60-80/hr as a ceiling, not the launch price.
University does not grant timely permission.	Medium	High	Secure written VC/administration sign-off before any public launch activity (14.2).
Theft, vandalism, or low civic-sense misuse.	Medium	High	ID-gated access, GPS tracking, refundable deposit; see Bounce/Vogo losses in 13.10.
Female users decline to use the service on safety grounds.	Medium	Medium	Standard-issue helmets, well-lit stand, women-only slot.
Demand drops during semester breaks/exams.	High	Medium	Keep fleet small; align growth-roadmap gates to the semester calendar.
Outsiders use campus infrastructure.	Medium	Medium	ID-card verification at sign-up.

Reference Points

Yulu — Institutional Distribution, Not Direct-Only Acquisition

Yulu's early pricing was simple and low-friction — roughly ₹10 per 30 minutes for bicycles and ₹10 per 10 minutes for e-scooters — low enough to be an easy trial decision, and its growth relied on institutional partners providing space and internal distribution rather than cold acquisition. Lesson: secure university buy-in early; it is a distribution channel, not just paperwork.

Bounce & Vogo — Scale Without Density Discipline Is Risky

Bounce scaled to roughly 25,000 scooters and around 120,000 rides a day pre-pandemic, and Vogo focused on transit-hub placement, but both experienced heavy theft/vandalism losses, and Bounce's revenue reportedly contracted by 83% during COVID-19. Lesson: start intentionally small and secure the fleet before chasing scale.

Ola Pedal at IIT Kanpur — A Direct Campus Precedent, and a Warning

Ola launched Ola Pedal with 600 cycles at IIT Kanpur in December 2017; the experiment reportedly did not last more than a few months. This is the closest precedent to the current proposal — its failure cautions against a large fleet without dedicated, hands-on student ownership of daily operations, which is why this plan proposes a small, founder-run pilot (13.1, 14.3) rather than a large third-party rollout.

UC Davis, California — Student Safety Data on Campus Micro-Mobility

A 2022 campus travel survey at UC Davis reported that nearly a quarter (22%) of students who rode an e-bike on campus had experienced an on-campus fall serious enough to require an ER or hospital visit — materially higher than for regular-bicycle riders. Not directly comparable to Indian road conditions, but a concrete campus data point supporting mandatory helmets and rider safety orientation as part of the MVP.

Zypp Electric — A Longer-Term Scalability Lens

Zypp Electric provides fleet-as-a-service EVs, batteries, and maintenance to delivery platforms on B2B contracts rather than direct consumer acquisition. Not a Phase 1 fit, but a useful reference if the venture later explores institutional/B2B routes beyond individual student rentals.

Implementation Plan

Phase 1 : Research, Validation & Permissions (Week 1-2)

Research & Validation

Autos and roughly 200 e-rickshaws already operate in Hisar, with some reportedly already plying informally on the GJUST campus itself — meaning the venture is entering a market with an existing, if informal, competitor rather than a blank slate. Rapido and Uber also have a presence in Hisar per survey feedback and public listings. This changes the framing from "create demand" to "capture demand that is currently being served badly"

- Re-validate the ₹20-50/trip pricing with 15-20 of the original survey respondents before locking the rate card.
- Map the existing informal e-rickshaw/auto routes on campus to avoid direct conflict at the same pickup points.

Permissions

- Day 1-2: submit a written permission/space request to the VC office / Estate Office — this has not started and is the critical-path item; nothing public-facing should happen before it is filed.
- Request specifically covers: stand space, charging point access, and permission to display signage/branding on campus.

Vehicle Selection

Selection criteria, in priority order: (1) sub-25 km/h category to avoid RTO registration/licensing overhead during the pilot; (2) availability of spare parts/service locally in Hisar rather than only in metro cities; (3) purchase or lease price that fits the ₹1,00,000 vehicle line in the funding plan

(16.2); (4) battery swappability, which was directly requested in survey feedback ("Heavy-Duty Build Quality, Swappable Batteries") and reduces downtime during the 4-6 PM peak.

Supplier Identification

- Local/regional EV dealerships in Hisar or nearby Rohtak/Chandigarh for new low-speed models.
- Refurbished/secondhand EV marketplaces for a lower per-unit cost, given the tight ₹1.5 lakh budget.
- Short-term B2B lease options (the Zypp Electric model, 13.10) as a fallback if outright purchase is not cost-effective for a pilot of this size.
- Local cycle shops in Hisar for the 2 standard bicycles — lower cost, no dependency on EV supply chains.

Phase 2 — Pilot Implementation (Week 3-6)

- Procure the fleet; apply branding/decals and ID-verification stickers.
- Set up Stand #1 between the hostel cluster and the market/academic block, per the survey-suggested location.
- Booking system: WhatsApp number / Google Form MVP — not a native mobile app. A custom app is deliberately deferred to Phase 3 since the survey data does not yet justify the development cost, and the manual flow is sufficient to test demand.
- Verification: physical student ID check at first sign-up, logged against a simple spreadsheet, addressing the outsider-use concern from the survey.
- Operations: vehicle rotation and charging scheduled outside the 12-2 PM / 4-6 PM / 6-8 PM peak windows identified in the survey
- Soft launch to the warm list of survey respondents; open the "first 100 riders free" window (13.6).

Phase 3 — Feedback & Iteration (Week 7-10)

- Weekly review against the KPI table
- Feature improvements prioritized directly from survey/pilot feedback: extend-ride-time option, per-minute billing test (replacing the flat per-trip rate if data supports it), and pre-booking — all explicitly requested in the open-text responses.
- Operational optimization: adjust stand hours and vehicle count per stand based on actual usage patterns, not the original estimate.
- Go/no-go decision on Phase 2 fleet expansion, gated on ≥60% utilization and ≥40% repeat usage .

Phase 4 — Scaling (Month 4+, Conditional)

- Additional campuses within Hisar/Haryana, larger fleet, and a second/third stand.
- Automation: GPS-based vehicle tracking, and eventually IoT-based lock/unlock replacing manual verification.
- Future technology: swappable battery stations (as requested in feedback) to reduce charging

downtime at scale.

Development Timeline & Milestones

Week	Milestone	Depends On
1	Permission request filed; vendor quotes finalized	Nothing — must start immediately
2	Pricing re-validated; coordinators recruited	Access to survey respondent contacts
3-4	Fleet procured, branded, stand set up	University granting stand space (permission)
5	Soft launch to warm list	Fleet + stand ready
6	Public launch; "first 100 free" window closes	Soft-launch feedback incorporated
7-10	Weekly iteration; feature improvements tested	6 weeks of usage data
10	Phase 2 go/no-go decision	KPI thresholds met

Revenue Model & Financial Projections

Revenue Streams

Per-Trip Rental Fee (Primary, Phase 1)

The core and only revenue stream in Phase 1. Priced in the ₹20-50/trip range validated against survey open-text feedback rather than the ₹60-80/hour closed-choice figure (13.2). Benefit: simple to explain and bill manually via WhatsApp; low barrier to a first-time user. Risk: revenue is capped by fleet size and cannot scale without more vehicles. Scalability: converts naturally into per-minute billing once a booking app exists (14.2).

Membership / Subscription (Phase 2+)

Aimed at the 71% of respondents who travel outside campus daily or 2-3 times a week (13.2) — a weekly or monthly pass would suit this group better than paying per trip. Benefit: predictable recurring revenue and higher retention. Risk: requires enough Phase 1 usage data to price a pass without under- or over-charging frequent users. Scalability: a natural second revenue layer once ridership is stable.

Advertising / Branding (Phase 3+)

Small local sponsors (stationery shops, cafes, coaching centres near the market corridor) could pay for stand or vehicle branding. Benefit: low-effort incremental revenue. Risk: negligible in Phase 1 at this fleet size, and is not included in the Phase 1 financial projections below. Scalability: grows with fleet visibility, not fleet size, so it is a genuinely different revenue lever from ridership.

University Partnership (In-Kind, Not Cash)

Framed here as a cost-reduction mechanism rather than a revenue stream — free or subsidized stand space and electricity access from GJUST directly reduces the recurring opex in 15.3, which has the same effect on net margin as new revenue without requiring a new payer.

Corporate Sponsorship (Future, Phase 4)

A long-term scalability option (e.g., a company sponsoring the fleet in exchange for branding across multiple campuses) but explicitly excluded from current projections since it depends on scale this venture does not yet have.

Pricing Model

Proposed structure: flat ₹20 for rides under 15 minutes, +₹5 per additional 5 minutes, capped around ₹40-50 for longer trips. This directly matches the open-text feedback ("rent should be 30-50, as auto takes 20") and the repeated request for per-minute rather than per-hour billing.

Option	Cost for a Typical Hostel ↔ Market Trip	Wait Time
Walking	Free	0 (but 15-20+ minutes on foot)
Auto	₹20-30 (per survey open text)	Long — the #1 complaint (67.2%)
Proposed e-scooter rental	₹19 flat (under 15 min)	Near-zero — vehicle staged at the stand

Positioned between walking (free but slow) and auto (comparable cost but with the wait-time problem the survey identifies as the top complaint) — the value proposition is speed and certainty, not being the cheapest option on paper.

Cost Assumptions (Phase 1, Monthly, Recurring Only)

One-time setup costs (vehicles, branding, deposits) are covered by the ₹1.5 lakh funding above are not repeated here.

Cost Item	Monthly (₹)	Basis
Electricity (2 e-scooters, ~1 charge/day)	800	Low-speed EV battery (~1.5-2 kWh) at ~₹15/full charge × 2 vehicles × 26 operating days, based on 2026 Indian domestic tariffs of ₹5-8/unit
Maintenance (tyres, brakes, minor repair)	2,000	Drawn from the ₹5,000 maintenance reserve in Section 16 over the pilot period
Referral credits	1,500	Capped pool from the marketing line in Section 16, concentrated in weeks 1-6
Tech / hosting / SIM	1,500	Amortized from the ₹10,000 tech line i above
Total recurring monthly opex	5,800	

Break-Even Analysis

At the recurring monthly opex of ₹5,800 and an average price of ₹25/trip, break-even requires roughly 232 rides/month, or about 2.2-2.3 rides/vehicle/day across a 4-vehicle fleet over 26 operating days. This is below the Phase 1 KPI target of ≥4 rides/vehicle/day (13.3), meaning the pilot is designed to be self-sustaining on recurring costs once usage stabilizes near target — though the Conservative case below shows it can still run a small loss if usage falls short.

Monthly Projections (Phase 1, 4 vehicles, 26 operating days)

Scenario	Rides/Vehicle/Day	Avg. Price/Trip	Monthly Revenue	Monthly Opex	Monthly Net
Conservative	2	₹20	₹4,160	₹5,800	−₹1,640
Expected	4	₹25	₹10,400	₹5,800	+₹4,600
Optimistic	6	₹30	₹18,720	₹5,800	+₹12,920

Directional Outlook Beyond Phase 1

Phase	Fleet	Expected Monthly Revenue	Expected Monthly Opex	Expected Monthly Net
1 — Pilot	4 vehicles	₹10,400	₹5,800	+₹4,600
2 — Validate	8 vehicles (if funded)	~₹20,000	~₹10,500	~+₹9,500
3 — Optimize	8 vehicles + booking app	~₹22,000	~₹11,500	~+₹10,500

Summary of Financial Assumptions

- 26 operating days/month, not 30 — accounts for maintenance days and disruption.
- No scenario assumes 100% fleet utilization.
- Electricity, maintenance, referral, and tech costs are the only recurring monthly costs modelled; one-time setup is funded entirely above.
- Insurance, treated as a largely one-time/annual cost , is not repeated in the monthly recurring cost table.
- All Phase 2+ figures are directional and explicitly flagged as needing re-validation.

Funding Requirement & Utilization Plan (₹1,50,000)

A single new low-speed e-scooter (unregistered, sub-25 km/h category) currently retails from roughly ₹58,000 to ₹71,000 ex-showroom in India. Buying two brand-new units would consume most of the ₹1.5 lakh budget and leave nothing for operations. The plan below uses 2 refurbished/negotiated-price e-scooters plus 2 standard bicycles to keep the pilot financially realistic.

Utilization Table

Category	₹	Why It Exists / MVP Relevance	Deferrable?
2 e-scooters (refurbished/fleet price) + 2 bicycles	1,00,000	Core pilot fleet — the MVP itself	No
Charging equipment	4,000	Keeps fleet available through the 4-6 PM peak window	Partially — 1 charger can rotate 2 EVs off-peak
Safety gear (helmets, locks)	6,000	Most-repeated open-text request in the survey	No
Insurance (basic, pilot period)	8,000	Liability safeguard for shared/student use — sub-25 km/h EVs are not legally required to carry motor insurance, but pilot-stage cover is a prudent safeguard	Yes, if a lower-cost group cover is found post-pilot-start
Registration/permission paperwork	1,000	Low-speed EVs need no RTO registration — this covers university admin process only	No
Branding (decals, signage, ID cards)	5,000	Stand visibility and ID-gated access control	Partially
Marketing (print, launch, referrals)	8,000	Funds the acquisition tactics in 13.6	Partially — referral pool can be capped lower
Tech (booking form/bot, hosting, SIM)	10,000	Supports the WhatsApp/Google-Form MVP flow (14.2)	Partially
Maintenance reserve	5,000	Directly protects the fleet-downtime KPI (15.3)	No
Emergency reserve	3,000	Buffer for quoted-vs-actual local price variance	No
Total	1,50,000		

Social / Economic Impact

Social Impact

Reduces campus carbon footprint — replaces petrol two-wheelers for short intra-campus commutes (hostel↔department↔library↔canteen), cutting emissions in a concentrated, high-traffic zone.

Solves last-mile mobility — most Indian university campuses (GJUS&T included) are sprawling; students without personal vehicles lose 15-20 mins walking between blocks. EV cycles/scooters cut this to minutes.

Inclusive access — affordable pay-per-ride/subscription model means even students without their own bike get equal mobility, reducing dependency on friends or costly autos.

Health & safety — encourages shift from unsafe shared rides/lift-sharing to dedicated, trackable transport; reduces campus traffic congestion near hostels.

Builds sustainability culture — positions the university as a green-campus pioneer, useful for

NAAC/NIRF green ranking parameters.

Economic Impact

Revenue model: subscription (monthly pass), pay-per-use (per km/per ride), and charging only fees for personal EV owners.

Job creation: station maintenance staff, charging attendants, app/backend management — student part-time roles

Cost savings for students: cheaper than autos/petrol scooters over a semester.

University gains: reduces need for internal shuttle subsidies; can be positioned as a revenue-sharing partnership with university facilities/transport dept.

Local EV ecosystem boost: partnerships with regional EV manufacturers/charging hardware startups (Haryana has growing EV manufacturing interest) — good local economic linkage story for judges

Future Scope & Scalability

Phase 1 (Pilot): 1-2 docking stations at high-footprint zones (main gate, hostels, academic block) with 3-4 cycles/scooters.

Phase 2: increase cycles/scooters. App-based booking + real-time availability + solar powered charging stations (sustainability angle = strong pitch differentiator).

Phase 3: Scale to other Haryana university campuses (NIT Kurukshetra, other state universities) via a B2B campus-mobility-as-a-service model.

Phase 4: Extend to nearby residential/commercial zones around campus (city-adjacent micro-mobility).

Tech roadmap: IoT-enabled locks, GPS tracking, predictive demand analytics, integration with student ID/app for seamless login.

Revenue diversification: advertising on stations, corporate sponsorships (EV brands), data insights sold to city planners on mobility patterns.

Conclusion

This isn't just a mobility fix — it's a scalable, revenue-generating sustainability model that starts hyper-local (one campus) and expands regionally. It solves a real, visible daily problem (campus commute), creates part-time student employment, cuts emissions, and builds a replicable SaaS-like franchise model for other campuses. With low capital risk in the pilot phase and a clear path to city/university-wide scale, it's positioned to deliver both social good and sustainable returns — exactly what seed investors look for in a campus-stage startup

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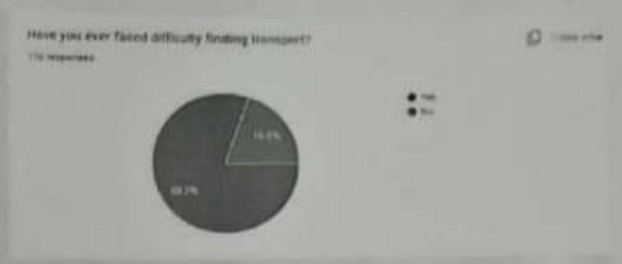
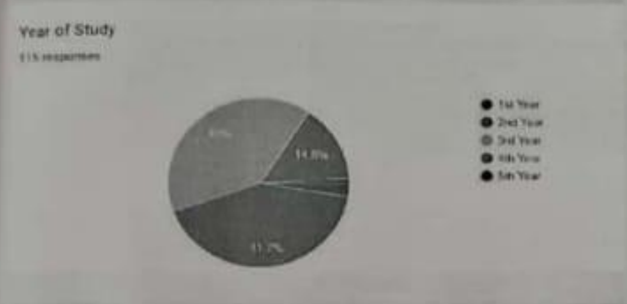
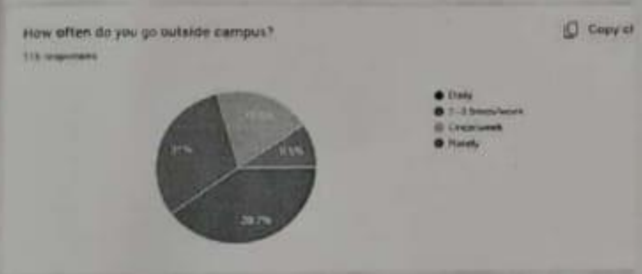
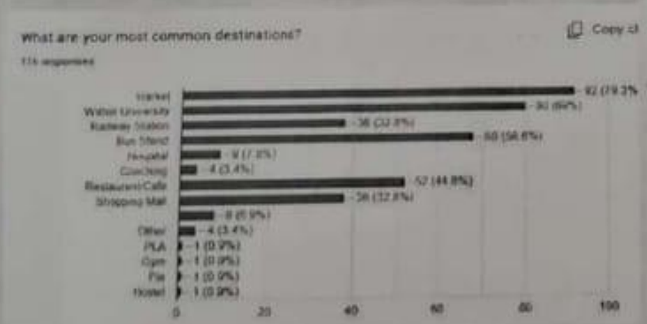
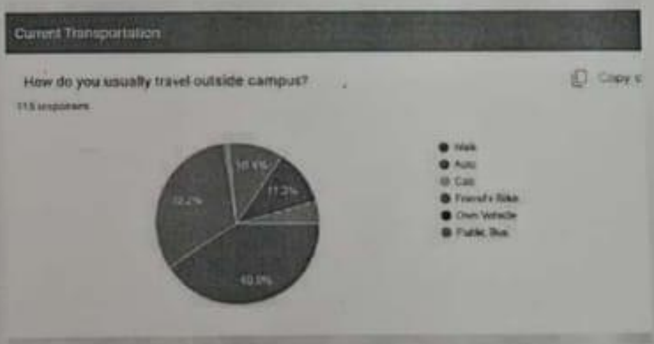
Supporting Survey











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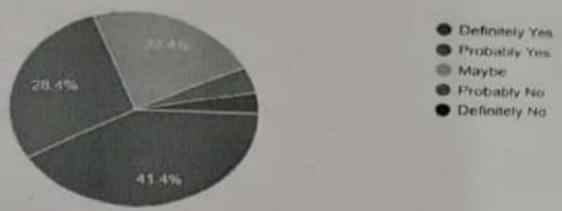
If electric scooters were available on campus for rent, would you use them?

116 responses

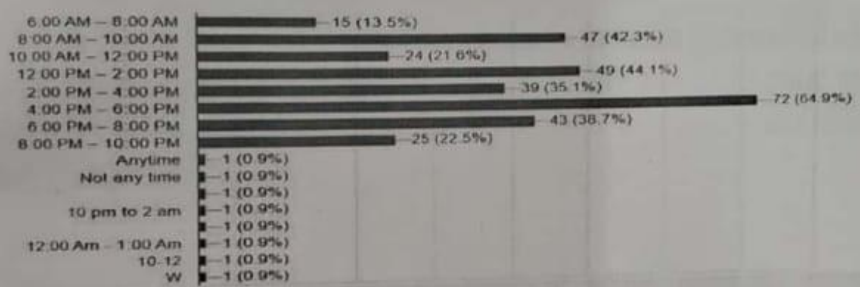
Copy chart



During which time do you usually need transportation?

111 responses

Copy chart



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Certificate

This is to certify that the project report titled **"Velo"** is an authentic and original work carried out by **"Sneha Singh"** and the startup team under the guidance and supervision of **"Dr. Sunil Verma"**

The team has sincerely worked on idea development, research, planning, and execution of the project. During the project period, the members demonstrated innovation, teamwork, creativity, and professional ethics.

To the best of my knowledge, this work has not been submitted elsewhere for any academic or commercial purpose. I appreciate the dedication and efforts of the team and recommend this project for further consideration under the **Seed Funding Program**.

10/7/26

Signature of Mentor


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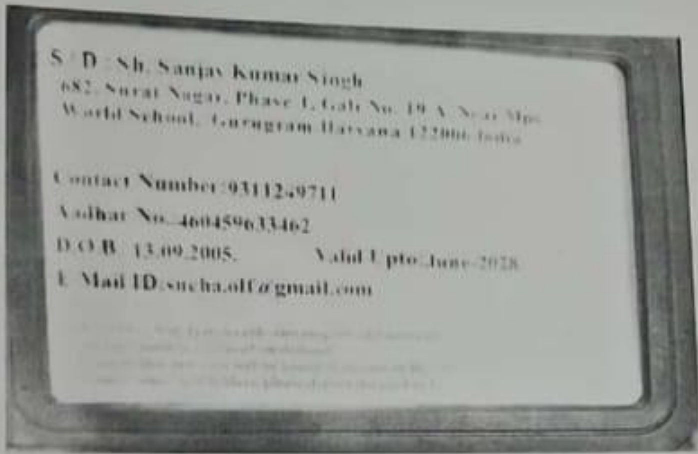
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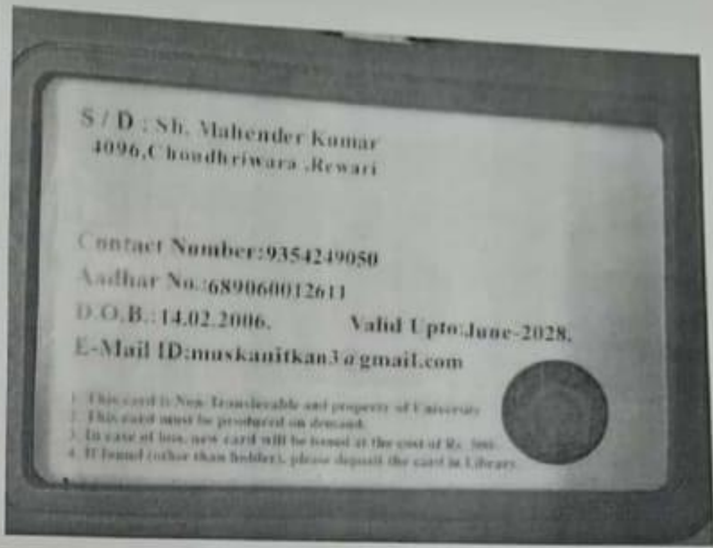
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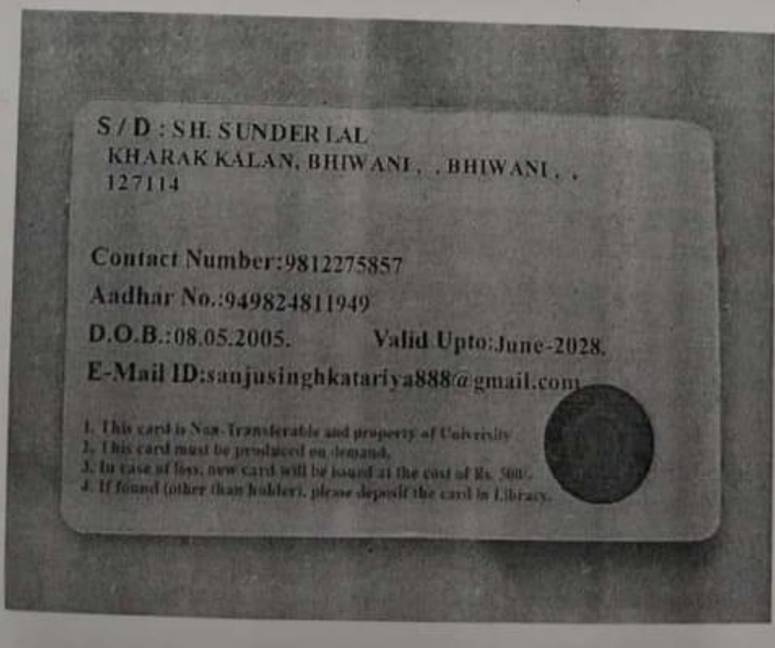

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